Hades Model Report

v0.8.4 GitHub

Hastie & Smith, et al. (2026)

MODEL CONFIGURATION

This report summarises the calibration of the model against the reference dataset, including parameter estimates and convergence diagnostics.

Mode	**Subalkaline**
Submode	**Calcalkaline**
Data suite	**./santa_maria.xlsx**
FeO/Fe^{2+}	**0.9**
Stages Run	Basic Intermediate

1. Results at a Glance

All of the mineral abundances and fractional crystallisation (FC) values for each phase are shown in the table below, as well as the resultant total fractions for the performed amount of fractional crystallisation over all phases.

The total melt fraction crystallised across all phases is 66.68 wt.%

	Basic	Intermediate	Total
Olivine	15.28		7.33
Clinopyroxene	0.69	15.82	8.56
Orthopyroxene	3.76	0.09	1.85
Plagioclase	58.37	51.18	54.63
Amphibole	17.06	10.34	13.57
K-feldspar		12.39	6.44
Biotite		0.78	0.41
Quartz		1.48	0.77
Magnetite	4.04	7.23	5.7
Ilmenite	0.8	0.69	0.74
FC%	32.0	51.0	

Table 1: Optimised mineral weights for the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

2. Parameter Configuration

The parameters below define the model as it was executed for this run.

	Primary	Primitive	Basic	Intermediate	Evolved
Plagioclase	3.4	3.4	3.4	3.4	3.4
Clinopyroxene	0.28	0.28	0.28	0.28	0.28
Orthopyroxene	0.29	0.29	0.29	0.29	0.29
Amphibole	0.37	0.37	0.37	0.37	0.37
K-feldspar	1.3	1.3	1.3	1.3	1.3
Biotite	1.0	1.0	1.0	1.0	1.0
Titanaugite	1.0	1.0	1.0	1.0	1.0
Nepheline	1.0	1.0	1.0	1.0	1.0
Leucite	1.0	1.0	1.0	1.0	1.0
Nosean/Hauyne	1.0	1.0	1.0	1.0	1.0
Alkali-amphibole	1.0	1.0	1.0	1.0	1.0

Table 2: Partition coefficients for the model phases.

3. Magma Evolution

Initial Compositions

	Sample	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
Primary									
Primitive									
Basic	samp. SM104	53.8	0.82	18.84	8.21	5.37	8.51	3.65	0.8
Intermediate	samp. S1120	58.46	0.83	17.84	7.33	2.97	6.94	4.36	1.27
Evolved	samp. S1004	65.54	0.39	16.89	4.5	1.6	4.25	5.16	1.68
End									

Table 3: Major oxide compositions of the model phases (wt.%).

Computed Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
Primary								
Primitive								
Basic	53.8	0.82	18.84	8.21	5.37	8.51	3.65	0.8
Intermediate	58.36	0.83	17.83	7.33	2.97	6.91	4.66	1.12
Evolved	65.48	0.45	17.06	4.37	1.6	4.31	5.09	1.63
End								

Table 4: Major oxide compositions of the model phases (wt.%).

4. Optimisation Results

Basic > Intermediate

BASIC

Population size	50
Iterations	500/500
Cost function	0.31
Trace analysis?	False

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	53.8	58.46	58.36	0.18
TiO2	0.82	0.83	0.83	0.43
Al2O3	18.84	17.84	17.83	0.01
Fe2O3	8.21	7.33	7.33	0.05
MgO	5.37	2.97	2.97	0.25
CaO	8.51	6.94	6.91	0.4
Na2O	3.65	4.36	4.66	6.9
K2O	0.8	1.27	1.12	11.78

Table 5: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	32.0
Olivine	15.28
Spinel	0.0
Clinopyroxene	0.69
Orthopyroxene	3.76
Plagioclase	58.37
Amphibole	17.06
K-feldspar	0.0
Biotite	0.0
Quartz	0.0
Magnetite	4.04
Ilmenite	0.8

Table 6: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
wt.%	44.05	0.8	21.02	10.34	10.25	11.93	1.49	0.12

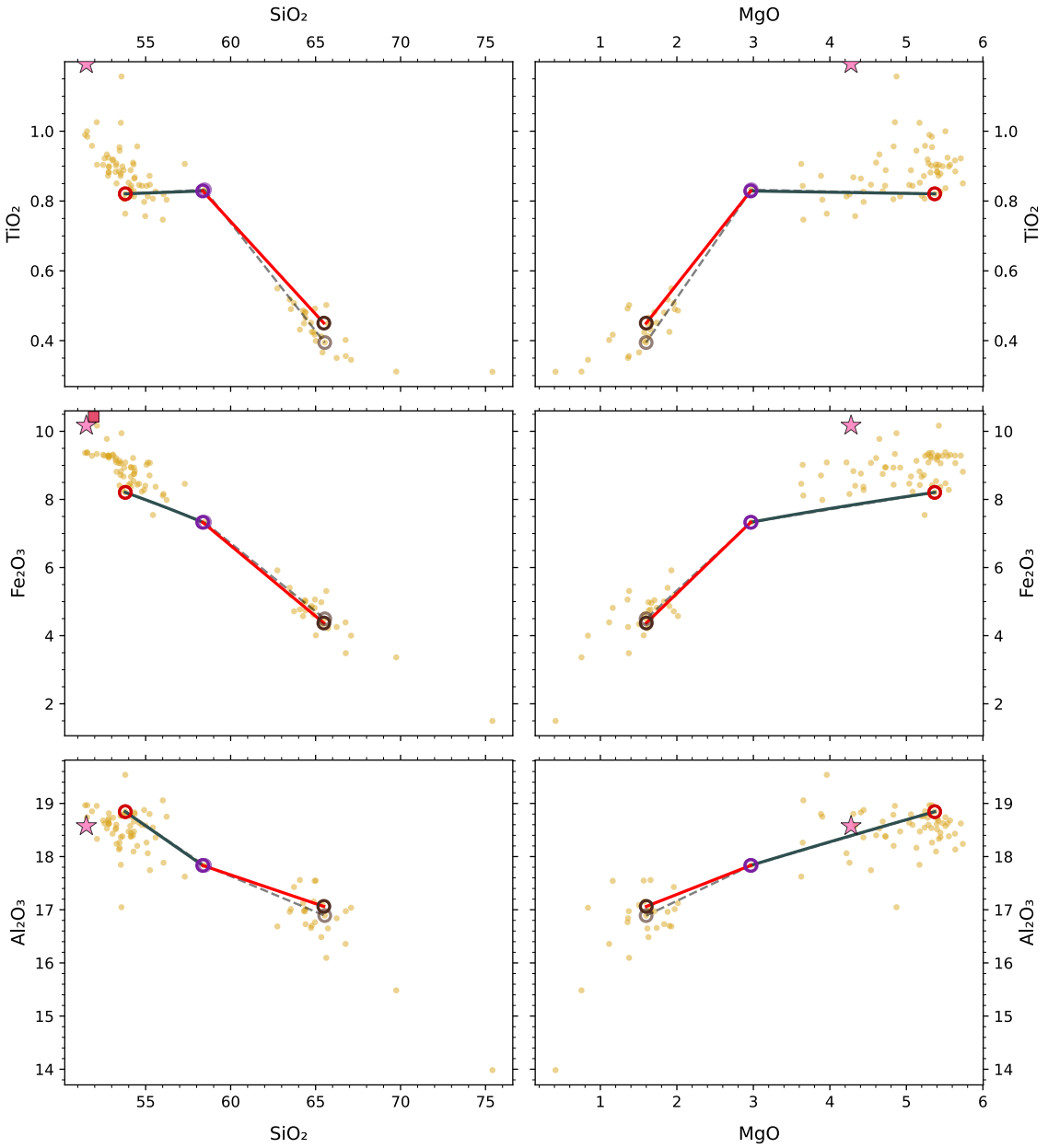
Table 7: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
olivine	38.34	0.03	0.38	19.14	41.56	0.55	0.0	0.0
spinel								
clinopyroxene	52.93	0.5	2.8	7.04	16.4	19.97	0.35	0.01
orthopyroxene	54.65	0.22	1.49	13.88	27.86	1.91	0.0	0.0
plagioclase	46.98	0.0	33.63	0.62	0.0	16.61	1.99	0.16
amphibole	48.91	1.09	7.35	13.17	16.03	11.37	1.92	0.17
kfeldspar								
biotite								
quartz								
magnetite	0.0	5.0	0.0	95.0	0.0	0.0	0.0	0.0
ilmenite	0.0	50.0	0.0	50.0	0.0	0.0	0.0	0.0

Table 8: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

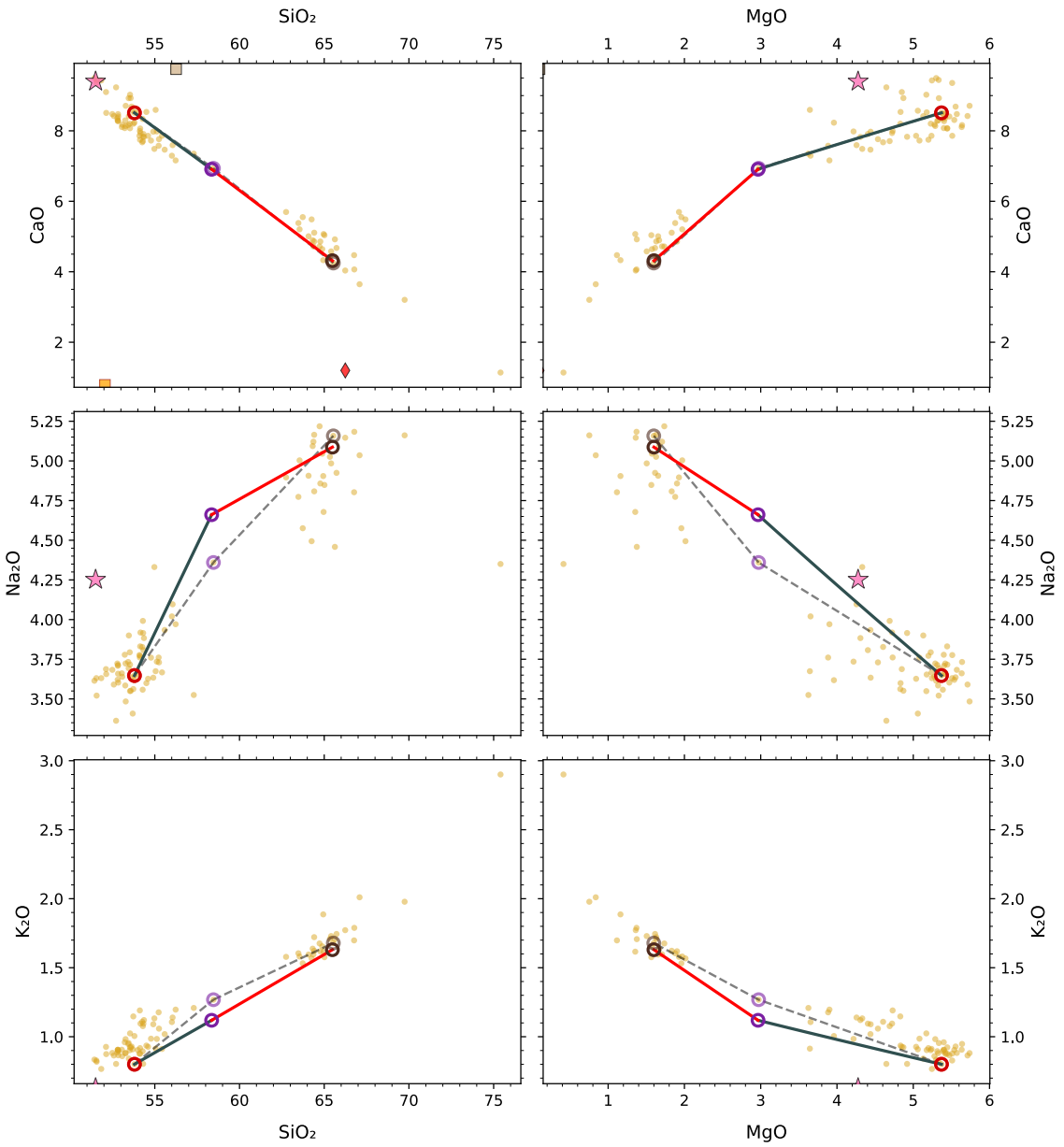


Figure 1: Bivariate plots of the data suite with the data-based and modelled LLODs. The Basic to Intermediate transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

Intermediate > Evolved

INTERMEDIATE

Population size	50
Iterations	500/500
Cost function	0.30
Trace analysis?	False

Magma Inspection

	Initial	Expected	Computed	% Error
SiO2	58.36	65.54	65.48	0.08
TiO2	0.83	0.39	0.45	14.15

	Initial	Expected	Computed	% Error
Al2O3	17.83	16.89	17.06	1.02
Fe2O3	7.33	4.5	4.37	2.74
MgO	2.97	1.6	1.6	0.13
CaO	6.91	4.25	4.31	1.57
Na2O	4.66	5.16	5.09	1.38
K2O	1.12	1.68	1.63	2.88

Table 9: Magma compositions for this phase (wt.% for oxides, ppm for trace elements).

Optimised Parameters

	Value
FC%	51.0
Olivine	0.0
Spinel	0.0
Clinopyroxene	15.82
Orthopyroxene	0.09
Plagioclase	51.18
Amphibole	10.34
K-feldspar	12.39
Biotite	0.78
Quartz	1.48
Magnetite	7.23
Ilmenite	0.69

Table 10: Optimised parameters for this phase. Fractional crystallisation (FC) is expressed as a percentage, and mineral weights are wt.%.

Bulk Crystallising Assemblage Composition

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
wt.%	51.51	1.19	18.57	10.17	4.28	9.4	4.25	0.63

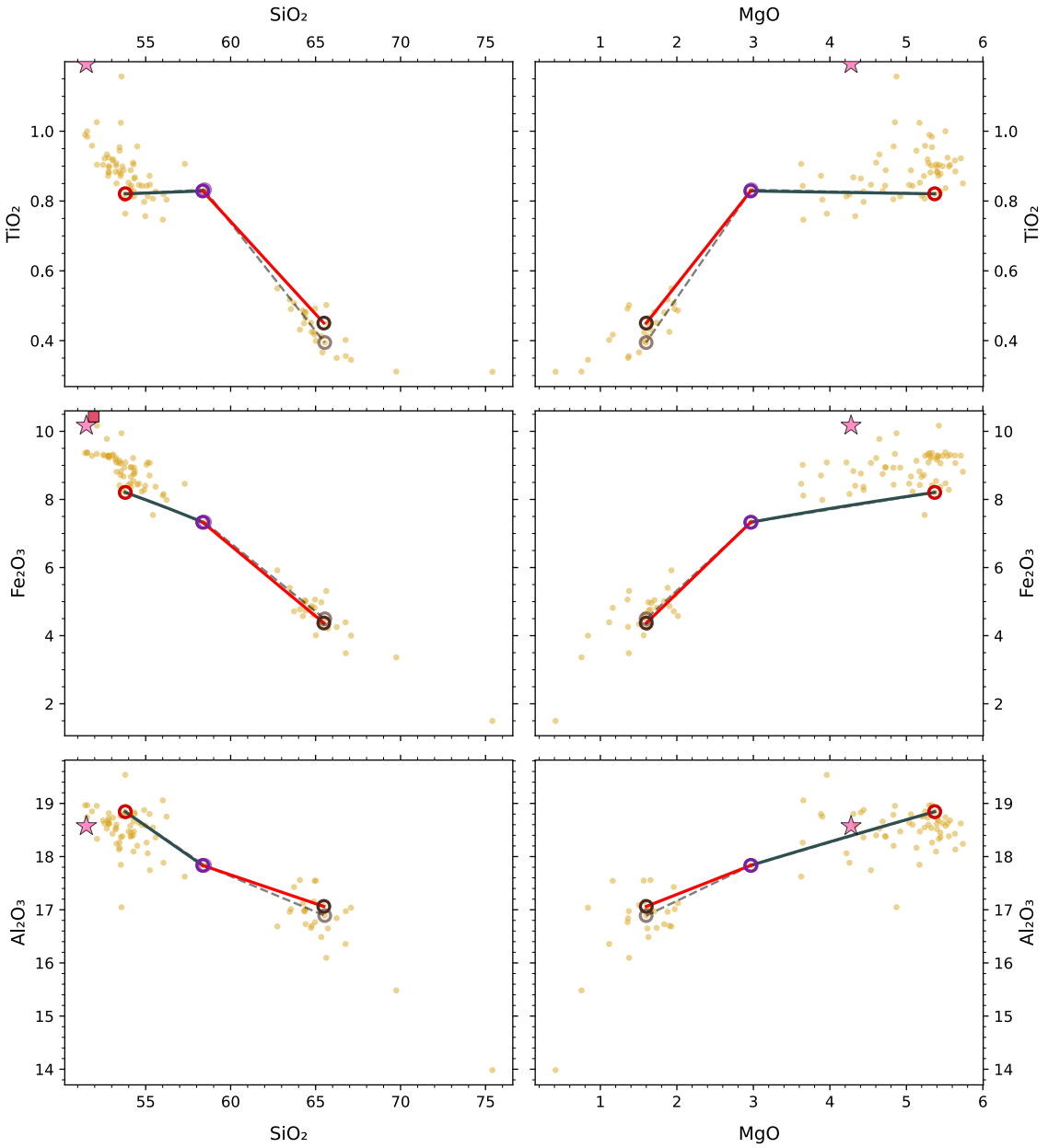
Table 11: Bulk mineralogy of the model phases (wt.%).

Cotectic Compositions

	SiO2	TiO2	Al2O3	Fe2O3	MgO	CaO	Na2O	K2O
olivine								
spinel								
clinopyroxene	51.94	0.2	1.63	10.43	15.4	19.79	0.61	0.0
orthopyroxene	52.06	0.1	0.79	20.92	24.95	0.79	0.39	0.0
plagioclase	56.26	0.0	27.83	0.33	0.1	9.75	5.53	0.2
amphibole	42.78	2.34	11.79	14.34	15.04	10.77	2.54	0.41
kfeldspar	66.24	0.08	20.38	0.18	0.1	1.2	8.5	3.31
biotite	37.08	3.43	17.29	23.02	9.3	0.38	0.21	9.27
quartz	100.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
magnetite	0.67	7.42	2.7	87.18	1.74	0.19	0.1	0.0
ilmenite	0.0	50.0	0.0	50.0	0.0	0.0	0.0	0.0

Table 12: Cotectic compositions of the model phases (wt.%). Blank rows indicate minerals that were not used in the optimisation for this phase.

Bivariate Plot



Bivariate plots continued overleaf.

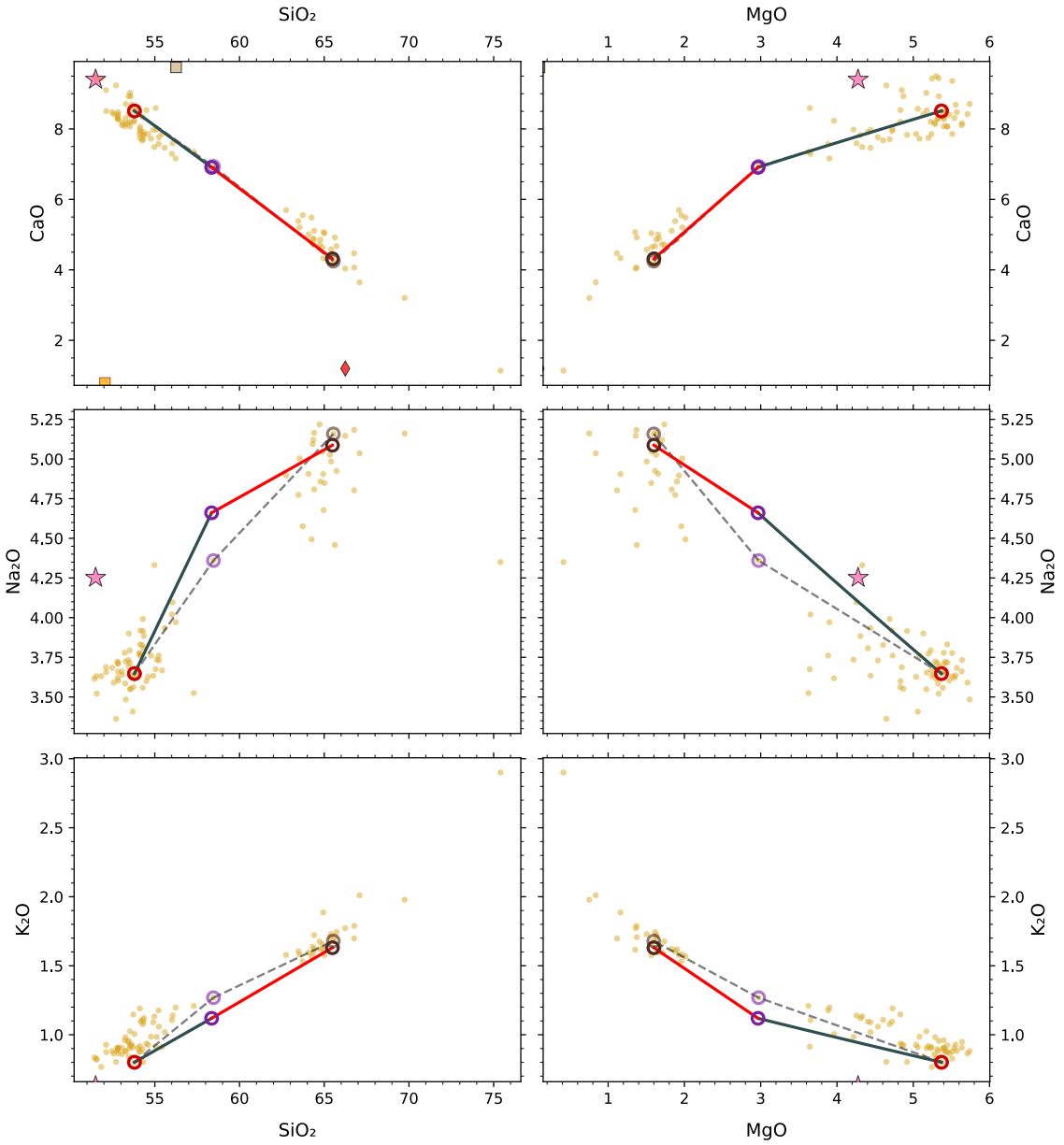


Figure 2: Bivariate plots of the data suite with the data-based and modelled LLODs. The Intermediate to Evolved transition is highlighted in red and mineral compositions are also displayed for this phase transition. See the legend at the end of the PDF report for the legend.

5. Figures

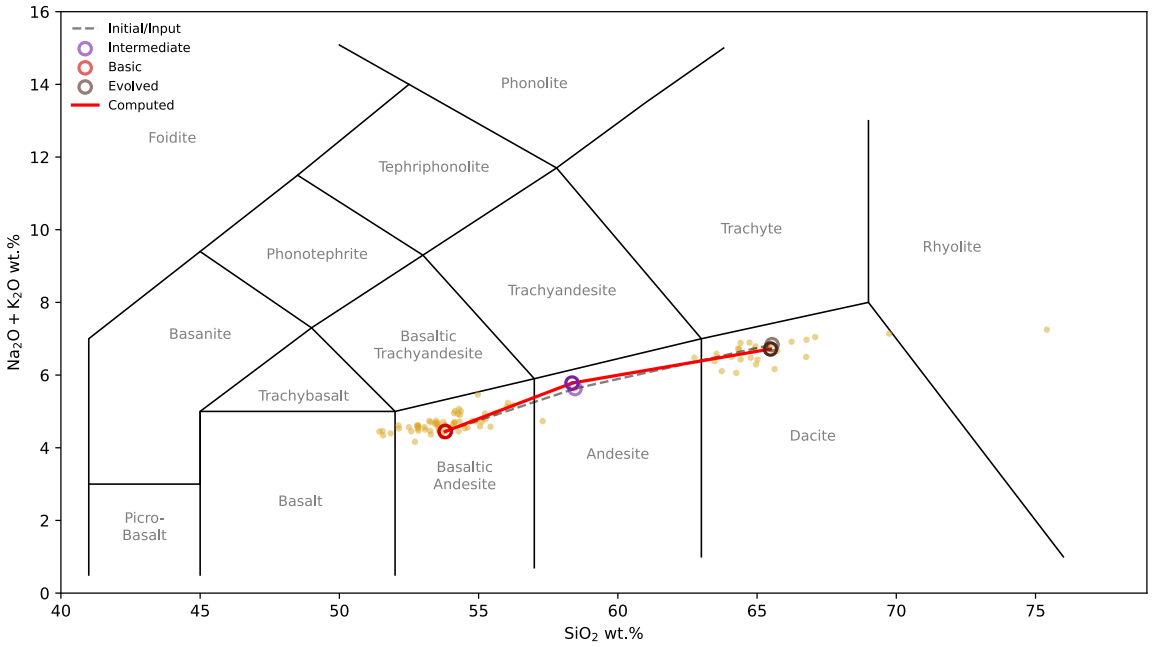


Figure 3: Total Alkali-Silica (TAS) diagram of the model phases.

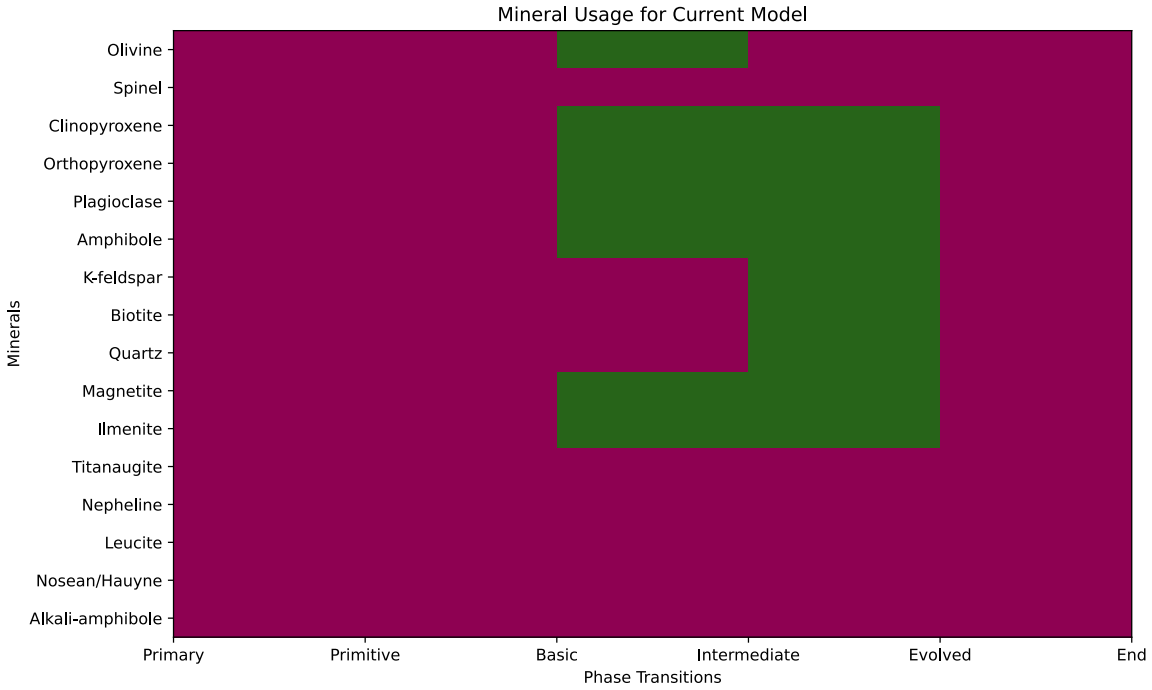


Figure 4: Mineral usage across the model phases.

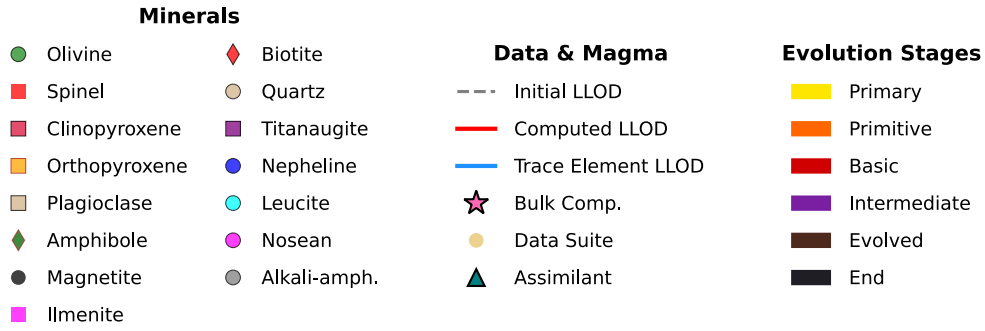


Figure 5: Legend for mineral symbols used in the figures.